VIJAY YADAV

PhD | Numerical Simulations | Scientific Machine Learning (SciML)

Indian ivijayyadav.github.io yadav.59@iitj.ac.in Publications 0009-0006-4903-8044 Jodhpur, IN

EDUCATION

PhD, Physics
Indian Institute of Technology Jodhpur
12/2021 – Ongoing Rajasthan, India

- Thesis title: Accelerating numerical simulations using machine learning.
- PhD advisor: Prabhat K. Jaiswal

Master of Science, Physics
National Institute of Technology Tiruchirappalli
08/2018 – 06/2020 Tamil Nadu, India

- Thesis title: CuS- rGO nanocomposites for gas sensing applications, Grade: 9.0
- Thesis advisor: S. Manivannan
- Final grade: 7.8

Bachelor of Science, Physics
University of Delhi
07/2015 – 07/2018 Delhi, India

- Dissertation title: Review on efficiency enhancement techniques for solar cell, Grade: 10.0
- Thesis advisor: Mukesh Kumar
- Final grade: 8.0

PUBLICATIONS

Published

1. V. Yadav, P. K. Jaiswal, R. Podgornik, and S. Kumari, "Charge Regulation Effect on Nanoparticle Interaction Mediated by a Polyelectrolyte," *Macromolecules* **59**, **11**, 6398-6409 (2026). [Selected for Front Cover]
2. V. Yadav, M. Priya, M. D. Shrimali, and P. K. Jaiswal, "Graph Neural Network for Prediction of Phase-Ordering Kinetics," *Chaos* **35**, 061101 (2025).
3. S. Chauhan, S. Mandal, V. Yadav, P. K. Jaiswal, M. Priya, and M. D. Shrimali, "Machine Learning Based Prediction of Phase Ordering Dynamics," *Chaos* **33**, 061103 (2023).

Submitted

1. V. Yadav, M. Priya, M. D. Shrimali, and P. K. Jaiswal, "Physics-Guided Convolutional Neural Network for Domain Growth Prediction in Systems with Conserved Kinetics," manuscript submitted (2026). [arXiv:2606.26128]

In Preparation

1. V. Yadav, S. Kumari, and P. K. Jaiswal, "Machine Learning Accelerated Simulation of Charge Regulation Effects," (2026).

PRESENTATIONS

07/2026: Flash talk at the Symposium on Complex Dynamics in Chemistry, Physics, and Biology (CDCPB 2026) on "Data-Driven Modelling of Phase-Ordering Kinetics."
03/2026: Oral talk (On-Demand) at the APS Global Physics Summit 2026 on "Machine Learning and GPU-Accelerated Modeling of Phase-Ordering Kinetics."
03/2026: Poster presentation (On-Demand) at the APS Global Physics Summit 2026 on "Charge Regulation Effects on Polyelectrolyte-Mediated Nanoparticle Interactions."
05/2023: Poster presentation at the CECAM workshop Mesoscale Modelling of Driven Disordered Materials: From Glasses to Active Matter, on "Study of Hessian Matrix Eigenvalues of the Glassy System Before and After Yield Stress and Its Prediction Using Random Matrix Theory."

TEACHING EXPERIENCE

2024–26: Delivered lectures and tutorials on Python programming fundamentals, NumPy, and data visualization for M.Sc. third-semester students (PHP 6070) at IIT Jodhpur, India.

2023–25: Delivered lectures and tutorials on Python programming fundamentals and algorithms for B.S. third-semester students (PHP 2010) at IIT Jodhpur, India.

AWARDS

2026: Article "Charge Regulation Effect on Nanoparticle Interaction Mediated by a Polyelectrolyte" featured on the front cover of *Macromolecules*, Vol. 59, Issue 11.

2024 – 2027: Senior Research Fellowship (SRF) from the Ministry of Education, India.

2022 – 2024: Junior Research Fellowship (JRF) from the Ministry of Education, India.

2017: Won first prize for the project "Drone Weather Station" during the Golden Jubilee function of SSN College, University of Delhi.

SKILLS

Programming Languages

Python | Mojo

Simulation Tools/Methods

LAMMPS | Finite Difference Method

Machine Learning Frameworks

PyTorch | JAX | Scikit-Learn

Visualization Tools

OVITO | Matplotlib

Additional Skills

SLURM Cluster | Git | LaTeX

LANGUAGES

English — Fluent

Hindi — Mother tongue